25901223 **Reg.No : .....**
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Value Addition Course (VAC) - MG4VACECO200 - SUSTAINABLE DEVELOPMENT STRATEGIES AND GOVERNANCE
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions Answer all questions Each question carries **1** marks

1. c) Environment–development conflicts
2. c) Our Common Future
3. d) WTO Agreement
4. b) Johan Rockström
5. c) Increased vulnerability
6. a) 1986
7. b) Petroleum
8. b) Fairness between present and future generations
9. b) United Nations Environment Programme
10. b) Housing development and environmental conservation
11. b) Strong family and peer support
12. d) Environment, health, and livelihoods
13. b) Nationally Determined Contributions (NDCs)
14. c) Evidence-based decision making
15. b) Environmental, Social, Governance

Part B

Short Answer Questions Answer any **10** questions Each question carries **2** marks

16. **Sustainability factors in rural industrial planning: -**While planning an industrial project in a rural area, sustainability requires balancing environmental, social, and economic aspects. Important factors include **environmental protection** (minimizing pollution and conserving biodiversity), **efficient use of natural resources** like water and energy, and **social impact** such as employment generation and avoiding displacement of local communities. Considering these factors ensures long-term viability and reduces negative externalities.

17. **Crisis or Catastrophe: -**The situation described is a **catastrophe**, as it involves large-scale destruction of infrastructure and overwhelms essential services like hospitals and emergency response systems. Unlike a crisis, which can be managed within existing capacities, a catastrophe exceeds the coping ability of institutions and requires external assistance.

18. **Transparency: -** Transparency refers to the **openness and accessibility of information in governance**, where decisions, policies, and processes are made visible to the public. It enables citizens to understand how decisions are made, promotes accountability, and reduces corruption by allowing public scrutiny.

19. **Rio+20:** Rio+20 refers to the **United Nations Conference on Sustainable Development held in 2012 in Rio de Janeiro**. It focused on promoting the **green economy** and strengthening institutional frameworks for sustainable development, building on earlier global environmental efforts.

20. **Rehabilitation in Disaster Management:** Rehabilitation is the **post-disaster phase** that focuses on restoring normal life by rebuilding infrastructure, reviving livelihoods, and providing social and psychological support to affected populations. It aims at long-term recovery and resilience building

21. **Three pillars of sustainable development:** Sustainable development is based on three pillars:
    - **Economic sustainability** (ensuring growth and efficient resource use)
    - **Social sustainability** (ensuring equity, justice, and inclusion)
    - **Environmental sustainability** (protecting natural resources and ecosystems)

    Balancing these pillars is essential for long-term development.

22. **Sustainability assessment:** Sustainability assessment is a **systematic process used to evaluate the environmental, social, and economic impacts** of a project, policy, or plan. It helps decision-makers ensure that development activities are sustainable and do not harm future generations.

23. **Social justice:** Social justice refers to the principle of **fair and equitable distribution of resources, opportunities, and rights** in society. It emphasizes reducing inequalities, ensuring inclusion, and protecting the rights of marginalized groups.

24. **2019 Climate Action Summit:** The 2019 Climate Action Summit aimed to **accelerate global efforts to combat climate change** by encouraging countries to strengthen their commitments to reducing greenhouse gas emissions and promoting sustainable development practices.

25. **Pressure-State-Response Framework:** The Pressure-State-Response (PSR) framework explains environmental issues through three stages:
    - **Pressure:** Human activities exert stress on the environment
    - **State:** The condition or quality of the environment changes

- **Response:** Society or governments take actions or policies to address the issue
  It helps in environmental policy analysis.

26. SDGs vs MDGs: The Sustainable Development Goals (SDGs) are broader and more comprehensive than the Millennium Development Goals (MDGs), covering economic, social, and environmental dimensions. However, despite improvements, challenges such as inequality, resource constraints, and implementation gaps remain, meaning SDGs have not fully overcome all limitations of MDGs.

27. Social-Ecological System: A Social-Ecological System (SES) refers to the **integrated and interdependent relationship between human societies and natural ecosystems**. Changes in one component affect the other, and understanding this interaction is essential for sustainable resource management.

28. Adaptive Capacity: Adaptive capacity is the **ability of individuals, communities, or systems to adjust to environmental, economic, or social changes**, cope with risks, and recover from shocks. Higher adaptive capacity leads to greater resilience.

Part C

Short Essay Questions Answer any **3** questions Each question carries **5** marks

29. Meaning and Concept of Governance: Governance refers to the **process by which decisions are made and implemented** in a society. It involves not only government institutions but also the participation of the private sector and civil society. Good governance is characterized by principles such as **transparency, accountability, participation, rule of law, and efficiency**.
In the context of sustainable development, governance plays a crucial role in designing and implementing policies that balance economic growth, social equity, and environmental protection. Effective governance ensures that resources are managed responsibly and that development benefits all sections of society.

30. Interrelationship between Biotic and Abiotic Factors: Biotic factors include all living organisms such as plants, animals, and microorganisms, while abiotic factors include non-living components like air, water, soil, sunlight, and temperature. These components are interdependent in an ecosystem.
For example, plants (biotic) depend on sunlight, water, and soil (abiotic) for growth. Animals depend on plants and environmental conditions for survival. Nutrient cycles and energy flows connect biotic and abiotic components, maintaining ecological balance. Any disturbance in one component affects the entire ecosystem.

31. Balancing Economic, Social, and Environmental Dimensions:
Sustainable development requires balancing three key dimensions:

- **Economic viability:** promoting growth, employment, and income generation
- **Social equity:** ensuring fairness, inclusion, and improved quality of life
- **Environmental protection:** conserving natural resources and ecosystems

If economic growth is pursued without environmental protection, it leads to resource depletion. Similarly, ignoring social equity leads to inequality and conflict. Therefore, a balanced approach ensures long-term sustainability and inclusive development.

32. Dimensions of Food Security: Food security has four main dimensions:

1. **Availability:** Adequate supply of food through production or imports
2. **Accessibility:** Economic and physical access to food for all individuals
3. **Utilization:** Proper use of food, including nutrition, hygiene, and health
4. **Stability:** Consistent availability and access over time without disruptions

All four dimensions must be ensured simultaneously to achieve true food security.

33. Community Participation and Capacity Building in Risk Management

Community participation plays a vital role in effective risk management. Local communities possess valuable knowledge about their environment and vulnerabilities. Their involvement improves disaster preparedness, response, and recovery.

Capacity building enhances skills, awareness, and resources, enabling communities to handle risks more effectively. It strengthens resilience and ensures that disaster management strategies are sustainable and locally relevant. Thus, community participation and capacity building are essential for reducing disaster risks.